

Multi-State Wave Computation via Dirichlet Characters: A Generalisation of the Möbius Interference Architecture

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Abstract

We generalise the Möbius wave computer architecture of [1] from binary to arbitrary q -state computation by replacing the Möbius function $\mu(n) \in \{-1, 0, +1\}$ with Dirichlet characters $\chi(n)$ of order q , whose values are q -th roots of unity. In the binary case, the Möbius function assigns two phases (0° and 180°) to signal paths, and Boolean logic emerges from two-phase wave interference. In the generalised case, a Dirichlet character of order q assigns q uniformly spaced phases ($2\pi k/q$ for $k = 0, \dots, q-1$), and q -state logic emerges from multi-phase interference.

We prove **multi-state functional completeness**: for any number of states q and any number of inputs k , every function $f : \{0, \dots, q-1\}^k \rightarrow \{0, \dots, q-1\}$ can be implemented as a character-weighted wave superposition. The proof uses the Fourier inversion theorem on finite abelian groups. We verify the theory computationally: all 27 single-input ternary functions (characters mod 7), all 256 single-input quaternary functions (characters mod 5), and large samples of multi-input gates all produce correct outputs.

We establish a **polynomial cascade theorem**: the output of one gate can feed into the next via polynomial evaluation in the complex amplitude, without intermediate phase detection or digital re-encoding. The polynomial degree required is $q-1$. For binary ($q=2$), this is degree 1 — cascading is **purely linear** and requires no nonlinear elements. For ternary ($q=3$), cascading requires a single quadratic nonlinearity (second-harmonic generation in optics). This result resolves the cascading problem — previously identified as the critical engineering bottleneck — for the binary case entirely, and reduces it to well-studied nonlinear optics for the multi-state case.

We discuss the physical implementation hierarchy: water waves support binary computation only (real-valued amplitude admits two phases), while electromagnetic waves natively support arbitrary q -state computation via complex amplitude. The generalised Euler product $L(s, \chi)$ is a native program in the multi-state architecture, connecting the computational framework to analytic number theory.

Keywords: Dirichlet characters, multi-valued logic, wave interference, Möbius function, functional completeness, L -functions, polynomial cascade, multi-phase computation

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1. Introduction

In [1], we presented a computational architecture — the “Natural Operating System” — in which Boolean logic emerges from the multiplicative structure of the integers. The architecture is governed by the master equation

$$y(S) = \sum_{n \in D} \mu(n) \cdot O_n \cdot [n \mid S] \cdot P_n(S)$$

where $\mu(n)$ is the Möbius function, O_n is a database output, $[n \mid S]$ is a divisibility-based activation indicator, and $P_n(S)$ tests a condition. We proved that the four standard Boolean gates (NOT, AND, OR, NAND) are implementable within this framework, establishing functional completeness for binary logic.

The Möbius signs $\mu(n) \in \{-1, +1\}$ (for squarefree n) implement the inclusion-exclusion principle: the OR gate uses three terms with coefficients $+1, +1, -1$, and the NOT gate operates by destructive interference between a constant reference and a phase-shifted input. These signs map directly to wave phases: $\mu(n) = +1$ corresponds to phase 0° (in-phase) and $\mu(n) = -1$ to phase 180° (anti-phase).

Water waves — real-valued displacements — support exactly two phases. But electromagnetic waves carry complex amplitude with continuous phase. This paper shows that by replacing $\mu(n)$ with Dirichlet characters $\chi(n)$ of arbitrary order q , the architecture generalises from binary to q -state computation. The mathematical foundation is Fourier analysis on finite abelian groups; the physical foundation is multi-phase wave interference.

1.1 Summary of Results

1. **Generalised master equation** (Section 3): The output is $y(x) = \sum_j \hat{F}_j \cdot \chi_j(x \bmod m)$, a weighted sum over Dirichlet characters.
2. **Multi-state functional completeness** (Section 5): Every function $\{0, \dots, q-1\}^k \rightarrow \{0, \dots, q-1\}$ is implementable.
3. **Polynomial cascade theorem** (Section 6): Gates cascade via degree- $(q-1)$ polynomial evaluation in the complex output, without intermediate detection. Binary cascading is linear.
4. **Computational verification** (Section 7): Exhaustive tests for $q = 2, 3, 4$ and large samples for $q = 5$.
5. **Physical hierarchy** (Section 8): Water $\rightarrow$ binary; EM $\rightarrow$ arbitrary q .
6. **Noise margin analysis** (Section 9): Quantitative bounds on tolerable per-source error.
7. **Generalised Euler product** (Section 10): $L(s, \chi)$ is a native program.

1.2 Relation to Prior Work

This paper extends [1] (the binary Möbius architecture) and draws on [2] (the logarithmic isomorphism) and [3] (the information spectrum). Multi-valued logic originates with Post [4] and Łukasiewicz [5]; ternary computing was explored by Brusentsov [6]. Wave-based computation is an active area in photonic computing [7, 8]. The connection between Dirichlet characters and multi-phase wave computation appears to be new.

1.3 Relation to Paper 2's Architecture

Papers [1] and the present work use compatible but structurally distinct architectures. Paper [1] uses **amplitude modulation**: fixed-phase sources with variable gates (physical barriers that open or close wave paths based on divisibility). The present paper uses **phase modulation**: variable-phase sources with no gates, where each source's phase is set by the character value $\chi_j(x)$ at the input. Both are physically realisable; the phase modulation approach is standard in phased array technology and telecommunications. For binary, either approach works. For multi-state, only the phase modulation (character-based) approach applies.

Paper [1]'s processor function $P_n(S)$ — a secondary condition test on the input — is absorbed into the character framework: the Fourier coefficients $\hat{F}_j$ already encode the complete logic function, including any conditional behaviour. The character evaluation $\chi_j(x)$ provides all routing information without a separate condition gate.

2. Mathematical Preliminaries

2.1 Dirichlet Characters

Definition 2.1. Let $q \geq 1$ be a positive integer. A *Dirichlet character* modulo q is a function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ satisfying: (1) $\chi(n) = 0$ if $\gcd(n, q) > 1$; (2) $\chi(n + q) = \chi(n)$ for all n ; (3) $\chi(mn) = \chi(m)\chi(n)$ for all m, n .

The characters mod q form a group of order $\varphi(q)$ under pointwise multiplication. When q is prime, this group is cyclic. A character of order d takes values that are d -th roots of unity.

2.2 Character Orthogonality

Theorem 2.2. For $a, b \in (\mathbb{Z}/q\mathbb{Z})^*$:

$$\sum_{\chi \bmod q} \chi(a) \overline{\chi(b)} = \begin{cases} \varphi(q) & \text{if } a \equiv b \pmod{q}, \\ 0 & \text{otherwise.} \end{cases}$$

2.3 Fourier Inversion

Theorem 2.3. Any function $f : (\mathbb{Z}/q\mathbb{Z})^* \rightarrow \mathbb{C}$ has a unique decomposition

$$f(a) = \sum_{\chi \bmod q} \hat{f}(\chi) \cdot \chi(a), \quad \hat{f}(\chi) = \frac{1}{\varphi(q)} \sum_{a \in (\mathbb{Z}/q\mathbb{Z})^*} f(a) \cdot \overline{\chi(a)}.$$

2.4 Recovery of the Möbius Function

The Möbius function $\mu(n)$ is not a Dirichlet character: it is not completely multiplicative ($\mu(p^2) = 0$ while $\chi(p^2) = \chi(p)^2$ for any character χ), and it is not periodic. The Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ is completely multiplicative but also not periodic; hence it too is not a Dirichlet character.

On the domain relevant to [1] — squarefree integers — $\mu(n) = \lambda(n) = (-1)^{\omega(n)}$, and both agree with the non-trivial real character of order 2 for each prime factor individually. Paper [1]'s binary architecture works because it restricts to squarefree addresses where this agreement holds.

The present paper’s character-based formulation is the cleaner generalisation; the Möbius-based formulation of [1] is the historically motivated special case.

2.5 Dirichlet L -Functions

For a character $\chi \bmod q$ and $\text{Re}(s) > 1$:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p)p^{-s}}.$$

3. The Generalised Master Equation

3.1 Definition

Definition 3.1. For a prime modulus m , with Dirichlet characters $\chi_0, \dots, \chi_{\varphi(m)-1} \bmod m$, and complex coefficients $\hat{F}_0, \dots, \hat{F}_{\varphi(m)-1}$ (the *database*), the **generalised master equation** computes:

$$y(x) = \sum_{j=0}^{\varphi(m)-1} \hat{F}_j \cdot \chi_j(x \bmod m)$$

where x is the encoded input.

Each term is a **wave source**: source j emits with complex amplitude $\hat{F}_j \cdot \chi_j(x \bmod m)$. The detector measures the total complex amplitude $y(x)$. The output state is:

$$\text{output} = \text{argmin}_{k \in \{0, \dots, q-1\}} \left| \arg(y(x)) - \frac{2\pi k}{q} \right|.$$

3.2 Physical Interpretation

Each wave source j has a fixed complex amplitude $\hat{F}_j$ (programmed from the Fourier coefficients of the target function) and a variable phase determined by $\chi_j(x \bmod m)$ (set by the input). The superposition at the detector is the Fourier synthesis. The output state is read from the phase sector of the total amplitude.

4. Input Encoding

4.1 Single-Input Encoding

A q -state input $A \in \{0, \dots, q-1\}$ is encoded as $A \mapsto g^A \bmod m$, where g is a generator of $(\mathbb{Z}/m\mathbb{Z})^*$ and m is chosen with $\varphi(m) \geq q$. The q residues $g^0, \dots, g^{q-1}$ are distinct.

Example 4.1 (Ternary). $q = 3$, $m = 7$, generator $g = 3$: $A = 0 \mapsto 1$, $A = 1 \mapsto 3$, $A = 2 \mapsto 2$. The order-3 character $\chi_2 \bmod 7$ maps these to distinct cube roots of unity, naturally distinguishing the three states.

4.2 Multi-Input Encoding

For k inputs, encode $(A_1, \dots, A_k) \mapsto \prod_{i=1}^k p_i^{A_i} \bmod m$ where $p_1, \dots, p_k$ are distinct primes.

Theorem 4.2 (Encoding existence). For any $q, k \geq 1$, there exists a prime m such that the q^k products $\prod_{i=1}^k p_i^{A_i}$ are distinct modulo m .

Proof. The products are distinct positive integers by the Fundamental Theorem of Arithmetic. Let $M = \prod_{i=1}^k p_i^{q-1}$ be the largest such product. Any prime $m > M$ makes all products distinct mod m (since they are positive integers less than m). By Bertrand's postulate, such a prime exists in $(M, 2M]$. $\square$

Remark 4.3 (Scaling). Using the first k primes, $M = \prod_{i=1}^k p_i^{q-1}$, so m (and hence $\varphi(m)$, the number of wave sources) grows exponentially in both q and k . This is worse than binary, where each input adds one prime with linear growth. The exponential scaling of multi-input encodings is the principal practical limitation of the framework.

5. Multi-State Functional Completeness

5.1 Statement and Proof

Theorem 5.1. For any $q \geq 2$ and $k \geq 1$, every function $f : \{0, \dots, q-1\}^k \rightarrow \{0, \dots, q-1\}$ is implementable by the generalised master equation.

Proof. (1) Choose primes $p_1, \dots, p_k$ and a prime modulus m as in Theorem 4.2 such that all q^k encoded residues are distinct in $(\mathbb{Z}/m\mathbb{Z})^*$.

(2) Define $F : (\mathbb{Z}/m\mathbb{Z})^* \rightarrow \mathbb{C}$ by $F(r) = \omega_q^{f(A_1, \dots, A_k)}$ when r is the encoding of $(A_1, \dots, A_k)$, and $F(r) = 0$ for non-encoding residues, where $\omega_q = e^{2\pi i/q}$.

(3) By Theorem 2.3, $F(a) = \sum_{j=0}^{\varphi(m)-1} \hat{F}_j \cdot \chi_j(a)$ with $\hat{F}_j = \frac{1}{\varphi(m)} \sum_a F(a) \cdot \overline{\chi_j(a)}$.

(4) Set the database coefficients to $\hat{F}_0, \dots, \hat{F}_{\varphi(m)-1}$. The generalised master equation then computes $y(x) = \sum_j \hat{F}_j \cdot \chi_j(x \bmod m) = F(x \bmod m) = \omega_q^{f(\text{inputs})}$.

(5) Since $|y(x)| = 1$ and $\arg(y(x)) = 2\pi f(\text{inputs})/q$, the output state is exactly $f(\text{inputs})$. $\square$

6. The Polynomial Cascade Theorem

6.1 The Cascading Problem

A multi-gate computation requires feeding the output of one gate into the input of the next. Gate 1 produces a complex amplitude $z_1 = \omega_q^k$ encoding state k . Gate 2 expects an encoded residue $r \in (\mathbb{Z}/m\mathbb{Z})^*$. The naive approach — detect the phase of z_1 , classify into state k , re-encode as $g^k \bmod m$, and re-emit — requires nonlinear detection and digital logic at each stage, which undermines the wave-based approach.

6.2 Polynomial Cascade

We show that this intermediate detection can be avoided entirely.

Theorem 6.1 (Polynomial cascade). Let gate 1 output $z_1 = \omega_q^k$ for input state k . The character value $\chi_j(g^k)$ needed by gate 2 is a polynomial of degree at most $q - 1$ in z_1 :

$$\chi_j(g^k) = \sum_{p=0}^{q-1} c_{j,p} \cdot z_1^p$$

for unique coefficients $c_{j,p} \in \mathbb{C}$. Therefore gate 2's output can be computed as:

$$y_2 = \sum_j \hat{F}_{2,j} \cdot \left(\sum_{p=0}^{q-1} c_{j,p} \cdot z_1^p \right) = \sum_{p=0}^{q-1} \left(\sum_j \hat{F}_{2,j} \cdot c_{j,p} \right) z_1^p$$

which is a degree- $(q - 1)$ polynomial in z_1 .

Proof. The values $\chi_j(g^k)$ for $k = 0, \dots, q - 1$ are determined by q constraints. The matrix $V_{kp} = (\omega_q^k)^p = \omega_q^{kp}$ for $k, p \in \{0, \dots, q - 1\}$ is the DFT matrix, which is invertible (its rows are distinct characters of $\mathbb{Z}/q\mathbb{Z}$, and characters of a finite group are linearly independent). Therefore there exist unique coefficients $c_{j,0}, \dots, c_{j,q-1}$ satisfying $\chi_j(g^k) = \sum_p c_{j,p} \cdot \omega_q^{kp}$ for all k . Since $z_1 = \omega_q^k$, we have $z_1^p = \omega_q^{kp}$, giving $\chi_j(g^k) = \sum_p c_{j,p} \cdot z_1^p$. $\square$

6.3 Binary Cascading is Linear

Corollary 6.2. For $q = 2$, the cascade polynomial has degree 1. Gate 2's output is:

$$y_2 = c_0 + c_1 \cdot z_1$$

This is a **linear** function of gate 1's output: a fixed complex constant plus a scaled copy of the input wave. Physically, this is just one more wave interfering with the output — no nonlinear element is required.

Corollary 6.3. Binary wave computation cascades via pure linear wave operations: superposition and fixed-amplitude scaling. No nonlinear optical elements, no detection, and no digital logic are needed at any intermediate stage.

This result means that arbitrarily deep binary circuits can be implemented in a single interference network — a mesh of linear wave paths with fixed complex weights. This is precisely the architecture of coherent photonic neural networks [7].

6.4 Ternary Cascading Requires Second-Harmonic Generation

For $q = 3$, the cascade polynomial has degree 2:

$$y_2 = c_0 + c_1 \cdot z_1 + c_2 \cdot z_1^2$$

The z_1^2 term requires squaring the complex amplitude. In optics, this is **second-harmonic generation** (SHG) — a well-studied process in nonlinear crystals (KDP, BBO, LiNbO₃). In electronics, it is analog multiplication.

More generally, q -state cascading requires generation of z_1^{q-1} , i.e., $(q - 1)$ -th harmonic generation. The nonlinearity order grows linearly with the number of states — a controlled, finite cost per cascade step.

6.5 Computational Verification of Cascading

We verified the polynomial cascade for the composition $\text{NEGATE} \circ \text{INCREMENT}$ in ternary ($q = 3$):

Input A	Gate 1 output z_1	Polynomial y_2	Phase	Detected	Expected
0	$-0.500 + 0.866i$	$-0.500 - 0.866i$	-120°	2	2
1	$-0.500 - 0.866i$	$-0.500 + 0.866i$	$+120^\circ$	1	1
2	1.000	1.000	0°	0	0

The polynomial cascade produces exact results with no intermediate detection.

7. Computational Verification

7.1 Summary of Tests

Test	States (q)	Modulus (m)	Functions tested	Correct	Method
Binary gates	2	3	4 (NOT, AND, OR, NAND)	4/4	Exhaustive
Ternary 1-input	3	7	27 (all)	27/27	Exhaustive
Ternary 2-input	3	19	500 (sampled)	500/500	Random
Quaternary 1-input	4	5	256 (all)	256/256	Exhaustive
Quinary 1-input	5	11	500 (sampled)	500/500	Random
Ternary cascade	3	7	1 (NEGATE INCREMENT)	3/3	Exhaustive

7.2 Named Ternary Gates

Gate	Definition	Wave sources used
IDENTITY	$A \mapsto A$	2 of 6
INCREMENT	$A \mapsto (A + 1) \bmod 3$	2 of 6
NEGATE	$A \mapsto (-A) \bmod 3$	2 of 6
CONSTANT(k)	$A \mapsto k$	2 of 6
MIN(A, B)	ternary AND	18 of 18
MAX(A, B)	ternary OR	18 of 18
ADD(A, B) mod 3	ternary addition	16 of 18

Gate	Definition	Wave sources used
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8. Physical Implementation Hierarchy

8.1 The Phase Constraint

Medium	Amplitude	Phases	Max q	Cascade nonlinearity
Water	Real	$\{0, \pi\}$	2	Linear (none needed)
Sound	Real	$\{0, \pi\}$	2	Linear
Electronic (I/Q)	Complex	$[0, 2\pi)$	Arbitrary	Analog polynomial
Optical (EM)	Complex	$[0, 2\pi)$	Arbitrary	Harmonic generation
Quantum photonic	Complex	$[0, 2\pi)$	Arbitrary	Gate-native

Water waves are limited to binary because surface displacement is real-valued: a crest is “positive” and a trough is “negative,” giving exactly two phases. Electromagnetic waves carry a complex electric field, supporting arbitrary phase. The ternary advantage (58.5% more information per measurement) and higher- q advantages are accessible only with complex-valued media.

8.2 Architecture Comparison

Paper [1] (binary Möbius) uses **fixed-phase sources + variable gates**: each wave source has a fixed phase determined by $\mu(n)$, and physical barriers open or close paths based on divisibility. The input determines *which* sources contribute.

The present paper uses **variable-phase sources + no gates**: each source’s phase is set to $\chi_j(x \bmod m)$ at the input. There are no barriers. The input determines *what phase* each source emits at.

Both are physically standard. Paper [1]’s approach maps to switched optical paths (Mach-Zehnder with on/off modulators). The present paper’s approach maps to phased arrays (sources with programmable phase shifters) — the same architecture used in 5G beamforming, radar, and coherent photonic processors.

9. Noise Margin Analysis

9.1 Phase Margin

The angular separation between adjacent output states is $2\pi/q$. A correct detection requires the measured phase to be within $\pm\pi/q$ of the true phase.

9.2 Error Tolerance

With $N = \varphi(m)$ sources and per-source amplitude error ε , the worst-case total output error is $N\varepsilon$ (by the triangle inequality). Detection succeeds when $N\varepsilon < \sin(\pi/q)$ (the projection of the error onto the decision boundary).

Maximum tolerable error per source:

$$\varepsilon_{\max} = \frac{\sin(\pi/q)}{\varphi(m)}$$

q	$\varphi(m)$	Margin ($^\circ$)	$\varepsilon_{\max}$	Required SNR (dB)
2	2	± 90	0.500	6.0
3	6	± 60	0.144	16.8
4	4	± 45	0.177	15.1
5	10	± 36	0.059	24.6
8	16	± 22.5	0.024	32.4

Binary is exceptionally robust: each source can err by up to 50% of its amplitude and the output is still correct. Ternary tolerates ~14% error per source. Beyond $q = 8$, the precision requirements become demanding.

9.3 Cascade Error Accumulation

In a cascade of d gates, each contributing error ε per source, the total accumulated error is bounded by $O(d \cdot N \cdot \varepsilon)$ for the detect-and-re-encode approach (errors reset at each stage) or potentially worse for the polynomial cascade (where errors compound multiplicatively through the polynomial). A detailed cascade error analysis is an open problem; the polynomial cascade's degree- $(q - 1)$ amplification of errors is the key concern.

10. The Generalised Euler Product as a Native Program

In [1], the partial Euler product $R(S) = \prod_{p \in S} (1 - 1/p^2)$ is a program in the binary master equation with $O_n = 1/n^2$. The same structure generalises: the Dirichlet L -function

$$L_S(2, \chi) = \prod_{p \in S} \frac{1}{1 - \chi(p)p^{-2}} = \sum_{\substack{n \text{ squarefree} \\ p|n \Rightarrow p \in S}} \frac{\chi(n)}{n^2}$$

is a program in the multi-state architecture. Each prime p is classified by χ into one of q phase groups. For the order-3 character $\chi_2 \bmod 7$:

Phase group	Residues mod 7	Primes (first few)
0° (state 0)	1, 6	13, 29, 41, 43
120° (state 1)	3, 4	3, 11, 17, 31

Phase group	Residues mod 7	Primes (first few)
240° (state 2)	2, 5	2, 5, 19, 23

Numerically, $L(2, \chi_2) \approx 0.799 - 0.102i$. This connects the wave architecture to the distribution of primes in arithmetic progressions (Dirichlet’s theorem) and to deeper structures in algebraic number theory.

11. Encoding Efficiency

The encoding efficiency $\eta = q/\varphi(m)$ measures the fraction of wave sources that encode useful states.

q	Min prime m	$\varphi(m)$	η	Information gain	Net advantage
2	3	2	1.000	1.000 bits	Baseline
3	5	4	0.750	1.585 bits	1.189×
4	5	4	1.000	2.000 bits	2.000×
5	7	6	0.833	2.322 bits	1.935×
8	11	10	0.800	3.000 bits	2.400×

The “net advantage” column is $\eta \cdot \log_2(q)$: the effective bits per wave source. Binary achieves 1.0 bit/source (baseline). Quaternary achieves 2.0 bits/source — a genuine doubling. Ternary’s net advantage of 1.189 bits/source is modest but nonzero. The optimal practical choice depends on the cascade architecture and the cost of nonlinear elements.

Remark 11.1 (Choice of modulus for ternary). The table shows that $m = 5$ with $\varphi(5) = 4$ suffices to distinguish 3 ternary states (efficiency $\eta = 0.75$). However, the ternary examples throughout this paper use $m = 7$ with $\varphi(7) = 6$, which has lower efficiency ($\eta = 0.5$) but provides a richer character group containing order-3 characters — whose values are exactly the cube roots of unity. This makes the ternary structure particularly transparent: the order-3 characters assign the three input states to phases $0^\circ, 120^\circ, 240^\circ$ directly. Using $m = 5$ would encode 3 states among 4 characters, all of which have order dividing 4, so the phase assignment would use 4th roots of unity rather than 3rd roots — functional but less natural. The choice of m involves a trade-off between encoding efficiency and algebraic elegance.

12. Discussion

12.1 What Holds and What Changed

The wave interference computer concept is **strengthened** by the revisions:

- The clean character-sum formulation (replacing the hand-wavy “divisibility activation at character addresses”) makes the physics transparent: each wave source has a programmable phase, and the interference computes the Fourier synthesis.

- The polynomial cascade theorem resolves the cascading problem — the critical engineering bottleneck — at least for binary (linear cascade) and makes it tractable for ternary (quadratic nonlinearity via SHG).
- The noise margin analysis confirms that binary is exceptionally robust and ternary is practical with standard optical/electronic precision.

12.2 Physical Limitations

1. **Multi-input scaling:** The number of wave sources grows exponentially with the number of inputs ($\varphi(m) \sim q^k$). This is fundamental to the encoding scheme, not an artifact. Binary avoids this because each bit is an independent prime, but the multi-state encoding couples all inputs into a single residue class.
2. **Cascade nonlinearity:** For $q > 2$, cascading requires nonlinear elements of degree $q - 1$. Second-harmonic generation (ternary) is well-studied, but higher-order nonlinearities are progressively harder and less efficient.
3. **Phase stability:** All wave sources must maintain stable phase relationships throughout computation. Phase drift between sources causes errors that accumulate across the cascade.
4. **Not Turing complete:** The framework establishes functional completeness for combinational logic. Sequential computation (memory, loops) requires feedback mechanisms not addressed here.

12.3 Open Problems

1. **Efficient multi-input encoding:** Can the exponential scaling of $\varphi(m)$ with k be avoided? Is there a q -ary analog of the binary “one prime per bit” scheme?
2. **Cascade error propagation:** How does error accumulate through polynomial cascades? The degree- $(q - 1)$ polynomial may amplify or suppress errors depending on the function; a full error analysis is needed.
3. **Optimal q :** Given a specific SNR and nonlinearity budget, what value of q maximises computational throughput? The net advantage table suggests $q = 4$ may be optimal for many settings.
4. **Connection to QAM:** The multi-phase output states are structurally identical to Phase-Shift Keying (PSK) constellations in telecommunications. The Fourier decomposition over characters is analogous to OFDM modulation. Are there deeper connections to be exploited?
5. **L -function computability:** Can the analytic properties of $L(s, \chi)$ be detected or computed within the wave architecture?

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Appendix A: Character Tables

A.1 Characters mod 5 (Quaternary, $q = 4$)

Generator $g = 2$. Encoding: state $0 \rightarrow 1$, state $1 \rightarrow 2$, state $2 \rightarrow 4$, state $3 \rightarrow 3$.

Let $i = e^{2\pi i/4} = \sqrt{-1}$.

n	$\chi_0(n)$	$\chi_1(n)$	$\chi_2(n)$	$\chi_3(n)$
0	0	0	0	0
1	1	1	1	1
2	1	i	-1	$-i$
3	1	$-i$	-1	i
4	1	-1	1	-1

χ_0 : trivial (order 1). χ_1 : order 4 (values are 4th roots of unity). χ_2 : order 2 (real-valued, Legendre-like). $\chi_3 = \overline{\chi_1}$: order 4.

A.2 Characters mod 7 (Ternary, $q = 3$)

Generator $g = 3$. Let $\omega = e^{2\pi i/6}$, $\zeta = e^{2\pi i/3}$.

n	χ_0	χ_1	χ_2	χ_3	χ_4	χ_5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	1	ω^5	ζ^2	-1	ζ	ω
3	1	ω	ζ	-1	ζ^2	ω^5
4	1	ω^2	ζ^2	1	ζ	ω^4
5	1	ω^4	ζ	1	ζ^2	ω^2
6	1	ω^3	1	-1	1	ω^3

Characters of order 3: χ_2 and $\chi_4 = \overline{\chi_2}$. These take values in $\{1, \zeta, \zeta^2\}$ — the cube roots of unity — and are the routing functions for ternary wave computation.

Appendix B: Notation

Symbol	Meaning
q	Number of computational states
m	Encoding modulus (prime)
χ_j	j -th Dirichlet character mod m
$\hat{F}_j$	Fourier coefficient / database entry for character j
ω_q	Primitive q -th root of unity: $e^{2\pi i/q}$
g	Generator of $(\mathbb{Z}/m\mathbb{Z})^*$
$\varphi(m)$	Euler totient: number of characters mod m
$L(s, \chi)$	Dirichlet L -function
η	Encoding efficiency: $q/\varphi(m)$
$\varepsilon_{\max}$	Maximum tolerable per-source error